

StartupHub: Pitch-to-Startup

AI-Powered Global Venture Ecosystem & Real-Time Pitch Matchmaking Platform

- **Presenter:** Imran Khan (GitHub: Imran8624)
- **Domain:** Full-Stack Engineering, Real-Time WebRTC & AI Integrations
- **Stack:** React 18, Vite, Tailwind CSS, Spring Boot 3, Java 17, H2 Database
- **Key Innovation:** Swipe Dealfow + 7-Slide Interactive Pitch Studio + WebRTC Video Rooms

The Early-Stage Startup Dilemma

Fundraising and recruitment remain archaic, fragmented, and excruciatingly slow.

- **Extreme Inefficiency:** Founders spend 6+ months emailing VCs with < 2% average response rates.
- **Static Pitch Decks:** Traditional PDF attachments lack dynamic financial models, founder voice, or interactivity.
- **VC Screening Overload:** Angel syndicates review hundreds of decks weekly with zero automated diligence.
- **Fragmented Workflow:** Startups juggle DocSend, Zoom, AngelList, and LinkedIn across disconnected silos.

Unified Venture Operating System

A single, real-time ecosystem connecting Founders, Investors, and Top Talent.

- **Tinder-Style Dealflow Swiper:** Native drag-physics with haptic audio feedback for rapid curation.
- **7-Slide Pitch Deck Studio:** Interactive problem, architecture moat, TAM/SAM/SOM, and unit economics.
- **Founder 60-Sec Elevator Pitch:** Audio waveform visualizer with live synchronized teleprompter subtitles.
- **1-on-1 Encrypted WebRTC Pitch Rooms:** Peer-to-peer HD video, screen sharing, and presentation timers.
- **Autonomous AI Diligence:** Instant valuation reality check, risk factor scoring, and 1-click Term Sheet generator.

System Architecture & Tech Stack

Decoupled, high-performance architecture built for global scale.

- **Client Tier:** React 18, Vite 5, Tailwind CSS, jsPDF, Web Audio API (Native tone synthesis).
- **API & Security:** Spring Boot 3.2.3, Java 17, Spring Security with stateless JWT validation.
- **Real-time Telemetry:** WebSocket (STOMP) protocol for signaling and real-time deal alerts.
- **Database Engine:** Persistent disk-backed H2 engine running in PostgreSQL compatibility mode.
- **DevOps & Edge:** Global edge CDN deployment on Vercel with single-page application rewrites.

AI Dealflow Swiper & Physics Engine

Intuitive gesture controls designed for high-velocity venture curation.

- **Physics-Based Dragging:** Real-time card translation (dx, dy) with dynamic angular rotation.
- **Visual Stamp Feedback:** Glowing overlays modulate opacity for INTERESTED, PASS, and SUPER-LIKE.
- **Stack Rewind / Undo:** Accidental passes can be rewound immediately via on-screen controls or Key Z.
- **Dual View Mode:** Instant toggle between immersive Card Swiper and Kanban/Grid Pipeline CRM.

7-Slide Interactive Pitch Deck Studio

Standardized, presentation-ready deck format for instant investor evaluation.

- **Slide 1 (Summary):** Target Ask, Valuation Cap, ARR velocity, and founding team pedigree.
- **Slide 2 & 3 (Problem & Moat):** Enterprise bottlenecks and proprietary sub-8ms latency architecture.
- **Slide 4 & 5 (Market & Traction):** TAM/SAM/SOM sizing (\$128B) and 6.8x LTV:CAC with 142% NRR.
- **Slide 6 & 7 (Cap Table & Ask):** ESOP pools, investor shares, and 18-month strategic milestone roadmap.
- **Export Capability:** One-click export of formal executive pitch decks into formatted presentation PDFs.

Founder 60-Second Elevator Audio Pitch

Restoring the founder voice and storytelling to remote pitch reviews.

- **Docked Audio Player:** Real-time frequency bars animated by mathematical waveform synthesis.
- **Synchronized Chapters:** Timestamps jump automatically between Hook, Breakthrough, Traction, and Ask.
- **Teleprompter Subtitles:** Live scrolling founder script ensures accessibility and fast scanning.
- **Direct Action Hooks:** Launch 1-on-1 pitch room, request NDA data room, or draft a Term Sheet.

Autonomous AI Due Diligence & Term Sheets

Automated venture intelligence that saves days of manual analyst review.

- **AI Scorecard (91-98/100):** Evaluates moat grade, capital efficiency, and burn multiple.
- **Comparable Multiples:** Benchmark valuations against public & private peers (Pinecone, Databricks, Stripe).
- **Partner Meeting Prep:** Generates tailored technical and pricing questions for investor syncs.
- **1-Click Term Sheet Builder:** Generates legal SAFE or Convertible Note agreements with instant PDF download.

1-on-1 WebRTC Pitch Room

Encrypted peer-to-peer video streaming without third-party meeting software.

- **Zero Media Server Lag:** Direct P2P video mesh utilizing browser RTCPeerConnection APIs.
- **Screen-Sharing Pipeline:** getDisplayMedia API powers seamless, crystal-clear slide presentations.
- **Signaling Infrastructure:** Spring Boot WebSocket handler coordinates ICE candidates and SDP handshakes.
- **Pitch Telemetry:** Includes integrated chat messaging and meeting duration stopwatches.

Talent Marketplace & ATS Optimizer

Connecting early-stage startups with vetted, mission-critical engineering talent.

- **Startup Roles Board:** Curated listings for AI/ML researchers, WebRTC leads, and systems engineers.
- **Automated ATS Match Score:** Calculates semantic keyword overlap between candidate and job profile.
- **Single-Column PDF Resume Generator:** Exports machine-readable PDFs built for Greenhouse, Lever, and Workday.
- **One-Click Apply:** Fast-track candidate pipeline directly visible in the startup management console.

Live Implementation & Deployment

Production-ready code running locally and committed to GitHub.

- **GitHub Repository:** <https://github.com/Imran8624/Pitch-to-Startup> (Branch: main).
- **Frontend Endpoint:** Localhost port 3000 | Vercel Edge deployment with SPA rewrites.
- **Backend Engine:** Localhost port 8080 | Java 17 / Spring Boot 3 with embedded Tomcat.
- **Database State:** 18 international startups pre-seeded across North America, Europe, and Asia-Pacific.

Project Summary & Future Horizons

Redefining the mechanics of global startup capital formation.

- **Summary:** StartupHub unifies deal discovery, interactive pitch decks, audio previews, video calls, and hiring.
- **Measurable Impact:** Eliminates 70% of initial fundraising screening lag for both founders and investors.
- **Future Horizon 1:** Smart contract escrow for automated SAFE execution upon term sheet agreement.
- **Future Horizon 2:** Live AI speech-to-pitch transcription with real-time objection handling prompts.
- Thank You! Questions & Live Demonstration.